

DSCC2010-4084

REDUCTION OF AN ELECTROCHEMISTRY-BASED LI-ION BATTERY HEALTH DEGRADATION MODEL VIA CONSTRAINT LINEARIZATION AND PADÉ APPROXIMATION

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ABSTRACT

This paper examines an electrochemistry-based model of health degradation via anode-side resistive film formation in Lithium-ion batteries. The paper makes this model more tractable and conducive to control design by making two main contributions to the literature. First, we adaptively solve the model's algebraic constraints using quasi-linearization. This improves the model's execution speed compared to solving the constraints via optimization. Second, we reduce the model's order by deriving a family of analytic Padé approximations to the model's spherical diffusion equations. The paper carefully compares these Padé approximations to other published methods for reducing spherical diffusion equations. Finally, the paper concludes with simulations of battery degradation that highlight the significant impact of the proposed model reduction approach on the battery model's overall accuracy and simulation speed.

NOMENCLATURE

a_j Specific area of the porous electrode j (1/m)
 c_1 Lithium concentration in the solid phase (mol/m³)
 c_2 Lithium concentration in the solution phase (mol/m³)
 $c_{1,j}^{\max}$ Maximum concentration of Lithium in the sphere
 $c_{1,j}^s$ Lithium concentration at the surface of sphere
 $D_{1,j}$ Diffusion coefficient of Lithium in the solid phase (m²/s)

D_2^{eff} Effective solution diffusion coefficient (m²/s)
 F Faraday's constant (C/mol)
 $i_{0,j}$ Main intercalation reaction current density (A/m²)
 $i_{0,sd}$ Side intercalation reaction current density (A/m²)
 i_{app} Applied current density (A/m²)
 J_1 Main intercalation volumetric current density (A/m³)
 J_{sd} Local volumetric current density for side reaction (A/m³)
 K_p Film conductivity (1/Ωm)
 k_j Rate constant of reaction, (A/m²) (mol/m³)^{1+α}
 L Length of the cell (m)
 M_p Molecular weight of side reaction product (kg/mol)
 $\bar{R}$ Universal gas constant (J/mol)
 R_{film} Film resistance at anode/solution interface (Ωm²)
 R_{SEI} Film resistance due to Solid Electrolyte Interface (Ωm²)
 R Radius of sphere (m)
 T Temperature (K)
 t^+ Transference Number
 $U_{ref,j}$ Local equilibrium potential for main reaction (V)
 U_{sd} Local equilibrium potential for side reaction (V)
 α_a Anodic transfer coefficients of electrochemical reaction

α_c	Cathodic transfer coefficients of electrochemical reaction
$\varepsilon_{1,j}$	Volume fraction of electrode j
$\varepsilon_{2,j}$	Volume fraction of solution
ϕ_1	Local potential of solid (V)
ϕ_2	Local potential of solution (V)
η_j	Local overpotential driving main intercalation reaction (V)
η_{sd}	Local overpotential driving side reaction (V)
κ^{eff}	Effective conductivity of solution (S/m)
κ_D	Diffusional conductivity of solution (S/m)
σ_j	Conductivity of electrode (S/m)
ρ_p	Density of active material (kg/m ³)
δ_{film}	Resistive film thickness, (m)

INTRODUCTION

This paper examines the problem of developing reduced, electrochemistry-based models of the dynamics of charging, discharging, and health degradation in Lithium-ion batteries. The overarching goal of the paper is to develop Lithium-ion battery models that satisfy two important but potentially conflicting objectives. First, the models must have the ability to accurately predict the performance and aging of Lithium-ion batteries in applications involving potentially complex and rapid charge/discharge cycles, e.g., hybrid vehicle applications. Second, the models must run with sufficient speed to enable battery system design, optimization, and control.

Several models have been used to monitor battery state of charge (SOC) and state of health (SOH) [1-4]. While these models are very desirable for control and estimation, they are unable to predict battery degradation. Predicting degradation for a wide variety of batteries requires an electrochemical model. One such model is provided by Doyle, Fuller, and Newman, and the addition of degradation is provided by Ramadass *et al* [5-8]. There are two major numerical difficulties with this electrochemical model. The first is the large number of state variables: a finite difference discretization of the model with M points along the width of the cell and N points in the pseudo-spherical direction has approximately $(2/3)*M*N$ state variables. The second challenge is the model's nonzero index, represented by approximately $(5/3)*M$ constraint equations, most of them involving a hyperbolic sine nonlinearity. This results in a large set of differential algebraic equations (DAEs). Ideally, one would like a model that: (i) runs quickly with a low number of state variables to enable optimal design and control studies, while (ii) still retaining the ability to accurately model complex, high-rate charge/discharge cycles.

Applying model reduction techniques to the above electrochemical battery model can bring it closer to the ideal speed and fidelity goals. Several reductions of this model are already presented in the literature. Some of these reductions pay special attention to the spherical diffusion submodel because it

appears repeatedly within the full battery model and contributes significantly to its state variable count. Reduction of the spherical diffusion submodel has been approached in a variety of different ways. Spherical diffusion dynamics have been approximated by parabolic and quartic profiles [9]. Proper orthogonal decomposition (POD) and residue grouping (RG) have also been used with great success [10,11]. The profile approximations are analytic but have not been generalized to higher orders. Both POD and RG are numerical in nature and POD requires, as input, a representative set of battery state trajectories.

In addition to the above studies focusing on spherical diffusion, researchers have also examined the problem of reducing the electrochemistry-based battery model as a whole. The single particle model and electrode-averaged model have been shown to be excellent at rates up to at least 1C, where "C" is defined as the current needed for fully charging/discharging a battery in 1 hour [4,12]. These models have the added advantage of not having to deal with large sets of algebraic constraints. POD has been used on the full battery model [10]. This furnished a reduced model capable of simulating constant high-rate charging and discharging. However, the model's performance increase was limited by the need to still solve a large set of algebraic constraints. A model reduction based on "volume averaging, approximation methods, and intuition" is presented in [13], and shown to be computationally efficient for lower charge/discharge rates.

This paper contributes to the above literature by providing two additional unique reductions to the above electrochemistry-based battery model. The first contribution is a quasi-linearized model of intercalation current that aids in efficiently solving the model's algebraic constraints. The second contribution is a family of analytic Padé approximations to spherical diffusion that substantially reduce the number of state variables present in the model while retaining a high level of accuracy.

The remainder of the paper introduces and discusses the full battery model, then shows how to efficiently solve the model's constraint equations via quasi-linearization and how to reduce its order substantially through a family of Padé approximations of the spherical diffusion equation. The paper then compares the Padé approximations to other methods used in the model reduction of spherical diffusion. Finally, the paper presents numerical simulations of the reduced battery model, and provides some conclusions.

THE LITHIUM-ION BATTERY MODEL

This section summarizes the Lithium-ion battery model originally developed by Doyle, Fuller, and Newman with a degradation component added by Ramadass *et al*. [5-7,12]. The section presents a brief summary of the main phenomena responsible for charge and health dynamics in Li-ion batteries, together with the equations modeling these dynamics.

Lithium-ion battery cells are devices that store electric energy by shuffling lithium ions between low and high potential energy states. Lithium ions have the lowest potential energy when they occupy *interstitial sites* within the solid crystalline structure of the cathode. Furthermore, Li-ions have the highest potential energy when they occupy interstitial sites within the solid crystalline structure of the anode. During charging, external current forces Li-ions to move from the cathode to the anode. This process can be conceptually divided into five steps. First, the Li-ions diffuse within the cathode's solid structure, towards the electrolyte. Second, the Li-ions travel by *intercalation* into the electrolyte solution. Third, the Li-ions diffuse through the electrolyte, across the cell's separator, and into the anode. Fourth, the Li-ions enter the anode's solid structure via intercalation. Finally, the Li-ions diffuse into the anode's interstitial sites, where the reduction-oxidation (REDOX) reaction is finally complete. During discharge, Li-ions trace this five-step process backwards. Specifically, they flow naturally from the anode to the cathode, thereby releasing Gibbs free energy and creating a useful current.

The above description summarizes the key dynamics governing charging and discharging in Li-ion batteries, but does not address battery health degradation. Lithium-ion batteries age and degrade due to a number of phenomena, including dendrite formation, carbon dissolution, electrolyte degradation, and electrode structure distortion [14,15]. This paper focuses on degradation via anode-side resistive film formation: a process known to have very significant impact on long-term battery health [7,12,14]. In this process, Li-ions travel by an irreversible *side intercalation reaction* into the *solid electrolyte interphase (SEI)* on the anode side of the battery cell. This causes the SEI film to grow, which increases the internal resistance of the battery and manifests itself externally in the form of power fade. This also results in permanent loss of cyclable lithium: a phenomenon that manifests itself externally in the form of capacity fade. The remainder of this section will present the equations governing both these degradation dynamics and the charge dynamics in Li-ion batteries [5-7, 12].

The diffusion of Li-ions within the electrolyte is governed by Fick's law of linear diffusion combined with an intercalation current density term, J , transferring Li-ions between the solution and solid:

$$\varepsilon_2 \frac{\partial c_2}{\partial t} = \nabla \cdot (D_2^{\text{eff}} \nabla c_2) + \frac{1-t^+}{F} J \quad (1)$$

The above intercalation current density, J , also acts as an input to the dynamics of Li-ion diffusion within the solid. This diffusion occurs at every point in the anode and cathode and can be modeled using a spherical, radially symmetric diffusion law as follows:

$$\frac{\partial c_{1,j}}{\partial t} = \frac{D_{1,j}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_{1,j}}{\partial r} \right) \quad (2)$$

The intercalation current density, J , consists of two components. The first component is a main intercalation reaction current density, J_1 , and the second component is the side intercalation current density, J_{sd} . The main intercalation current density is driven predominantly by potential differences between the solid and electrolyte solution, and governed by the well-known Butler-Volmer equation:

$$J_1 = a_j i_{0,j} \left[\exp \left(\frac{\alpha_{a,j} F}{RT} \eta_j \right) - \exp \left(-\frac{\alpha_{c,j} F}{RT} \eta_j \right) \right], \quad j = n, p \quad (3)$$

$$i_{0,j} = k_j (c_{1,j}^{\text{max}} - c_{1,j}^S)^{\alpha_{a,j}} (c_{1,j}^S)^{\alpha_{c,j}} (c_2)^{a_{a,j}}, \quad j = n, p \quad (4)$$

The overpotentials in the above equations equal the difference between the solid and solution potentials minus the reference potentials for the main intercalation reaction, which in turn depend on the local state of charge. In other words, the overpotentials are given by:

$$\eta_p = \phi_1 - \phi_2 - U_{\text{ref},p} \quad (5)$$

$$\eta_n = \phi_1 - \phi_2 - U_{\text{ref},n} - \frac{J}{a_n} R_{\text{film}} \quad (6)$$

Since the above potentials/overpotentials can change much faster than the Li-ion concentrations, they are typically assumed to respond instantaneously. This assumption is valid to approximately 100 Hertz [16]. The solid potential is governed by Ohm's law with a term governing the charge transfer due to intercalation:

$$\nabla \cdot (\sigma_j^{\text{eff}} \nabla \phi_{1,j}) - J = 0, \quad j = n, p \quad (7)$$

Similarly, the solution potential is governed by Ohm's law, intercalation current density, and the charge carried by the Li-ions in solution (also known as concentration polarization):

$$\nabla \cdot (\kappa^{\text{eff}} \nabla \phi_2) + J + \nabla \cdot (\kappa_D \nabla \ln(c_2)) = 0 \quad (8)$$

The above system of equations governs the dynamics of charging and discharging in the Li-ion cell. From a dynamic systems perspective, it is a system of *differential algebraic equations (DAEs)*, where the differential equations govern the diffusion dynamics and the algebraic constraints constrain the potentials and intercalation current accordingly. A degradation component can be added to this model by assuming that film growth in the anode is governed by an electrochemically driven irreversible side reaction. This side reaction is itself governed by a side reaction overpotential as follows:

$$\eta_{sd} = \phi_1 - \phi_2 - U_{ref, sd} - \frac{J}{a_n} R_{film} \quad (9)$$

$$J_{sd} = -i_{0, sd} a_n \exp\left(-\frac{\alpha_{c,n} F}{RT} \eta_{sd}\right) \quad (10)$$

The above side reaction creates a resistive film at a rate proportional to the side reaction current density, i.e.,

$$\frac{\partial \delta_{film}}{\partial t} = -\frac{J_{sd} M_p}{a_n \rho_p F} \quad (11)$$

The above resistive film adds to the internal resistance of the anode, thereby negatively affecting battery performance:

$$R_{film} = R_{SEI} + \frac{\delta_{film}}{K_p} \quad (12)$$

The above battery health degradation model has the advantage of being electrochemistry-based and therefore predictive in nature. It captures several important phenomenon of battery degradation. It captures power fade through the added resistance. Capacity fade is accounted for through J_{sd} consuming lithium and also the increased resistance. Additionally the model captures SOC dependent rates of degradation through η_{sd} which is affected by the difference in local SOC to the U_{sd} the equilibrium potential of the side reaction. However, it suffers from two main numerical challenges. First, Equations (3-8) create an algebraic loop within the model, thereby causing the model to have a nonzero index. This increases the computational cost of simulation the model considerably. Second, the fact that the spherical diffusion equation, Equation (2), must be satisfied at every point in the anode and cathode creates a need for potentially very large numbers of state variables just for the purpose of calculating the solid concentration at every discretized point along the battery electrodes. The literature recognizes these two challenges and presents some methods for addressing them. The main contribution of this paper is its contribution of constraint quasi-linearization and Padé approximation of spherical diffusion as two unique methods for addressing these challenges.

QUASI-LINEARIZATION APPROACH FOR SOLVING BATTERY ALGEBRAIC CONSTRAINTS

Equations (3-8) in the above electrochemistry-based battery model form an algebraic loop. To illustrate this, consider a Li-ion battery whose solid and solution concentrations and potentials are known at a particular instant in time. For this battery, one can use Equations (3-6) to determine the main and side reaction intercalation currents. Adding these two intercalation currents gives the total intercalation current, which in turn governs the solid and solution potentials through Equations (7-8). This algebraic loop, together with the concentration differential equations, Equations (1-2), forms a set of differential algebraic

equations (DAEs) governing the battery's charge and health dynamics. To simulate the battery, one must obtain consistent solutions to the algebraic constraints at every time step, in addition to using the concentration differential equations to propagate the battery dynamics in time.

The problem of obtaining consistent solutions to the algebraic constraints in a set of DAEs, at every integration time step, is essentially a root finding problem. One may therefore attempt to solve it using a number of different root-finding methods, including the Newton-Raphson method, fixed-point iteration, and optimization methods. These methods differ in numerical complexity and cost, but they all contribute significantly to the cost of simulating the full battery model numerically. Consider, for instance, the following iterative scheme for solving the battery model's algebraic constraints. Suppose one starts from known initial conditions for the solid and solution concentrations, plus initial guesses for the main and side reaction intercalation currents, J_I and J_{sd} . One can therefore add J_I and J_{sd} to obtain an initial guess for the total intercalation current density, J . Using Equations (7-8), one can obtain the solid and solution potentials given this J . The total squared difference between these solid and solution potentials and the assumed solid/solution potentials can be defined as a quadratic error function. Finally, an optimization routine such as sequential quadratic programming can be used to minimize this error function, thereby solving the battery model's algebraic constraints. The authors have used this approach to simulate the dynamics of a Li-ion cell successfully. However, the resulting simulation speeds were too slow for system identification, optimization, and control studies. We therefore propose an alternative approach that significantly reduces the computational cost of solving the battery model's algebraic constraints.

The proposed approach for solving the battery model's constraint equations relies on two important insights. First, aside from the Butler-Volmer (BV) expressions for the main and side reaction intercalation current densities, Eqs. (3,10), all other constraint equations in the battery model are linear. Second, while the BV equation itself is nonlinear, this nonlinearity tends to produce very large changes in intercalation current densities for small increases in overpotential. This, in turn, generally forces the overpotentials to remain within some reasonable bounds over which the Butler-Volmer equation can be *quasi-linearized*. Quasi-linearization, in this context, is the process of linearizing the Butler-Volmer equation at every integration time step. This approximates the algebraic loop in the battery model by a different set of linear equations at every integration time step. As the remainder of the paper will show, this approximation – together with the Padé approximation of spherical diffusion dynamics – increases overall model execution speed by 2-3 orders of magnitude with only a minimal loss of fidelity.

We begin with linearizing the BV kinetic expression for the main intercalation reaction, Eq. (3). Assuming the anodic and cathodic transfer coefficients are equal for the electrochemical

reactions in the anode and the cathode, i.e. $\alpha_{a,j} = \alpha_{c,j} = \alpha_j, j = n, p$ (which holds true for the simulations in this paper), the BV expression for the main intercalation reaction, Eq. (3), can be rewritten as:

$$J_{1,j} = \mu_j \sinh(\lambda_j \eta_j), \quad j = n, p \quad (13)$$

$$\mu_j = 2a_j i_{0,j}, \quad \lambda_j = \frac{\alpha_j F}{RT} \quad (14)$$

Note that $J_{1,j}$ becomes a vector of length j (the number of discretized points along electrode j) after applying the finite differences approximation to the system equations.

The first order Taylor series expansion around the operating point $\eta_{j,0}$ can be written as:

$$J_{1,j} = V_j + M_j (\eta_j - \eta_{j,0}) \quad (15)$$

$$V_j = \mu_j \sinh(\lambda_j \eta_{j,0}); \quad M_j = \text{diag}[\mu_j \lambda_j \cosh(\lambda_j \eta_{j,0})] \quad (16)$$

Convenience variables are now made use of:

$$\Lambda_n = I_n + \frac{R_{film}}{a_n} M_n; \quad \Theta_n = \frac{R_{film}}{a_n} M_n \quad (17)$$

$$\Gamma_j = M_j; \quad \psi_j = V_j - M_j (U_{ref,j} + \eta_{j,0})$$

Now we have matrix equations involving J_I and J_{sd} .

$$\Lambda_n J_{1,n} + \Theta_n J_{sd} = \Gamma_n (\phi_{1,n} - \phi_{2,n}) + \psi_n \quad (18)$$

$$J_{1,p} = \Gamma_p (\phi_{1,p} - \phi_{2,p}) + \psi_p \quad (19)$$

where $\Lambda_n, \Theta_n, \Gamma_n \in \mathbb{R}^{n \times n}$, $\Gamma_p \in \mathbb{R}^{p \times p}$ (with n and p being the number of discretized points along the negative and the positive electrode, respectively), $\psi_n \in \mathbb{R}^n$ and $\psi_p \in \mathbb{R}^p$. Eqs. (13) and (14) differ because side reaction film formation is only occurs at the anode.

Similar to the main reaction, the linearized form of the side reaction intercalation current density, Eq. (10), can be written in the form of a linear matrix equation as well:

$$\Lambda_{sd} J_{sd} + \Theta_{sd} J_{1,n} = \Gamma_{sd} (\phi_{1,n} - \phi_{2,n}) + \psi_{sd} \quad (20)$$

where $\Lambda_{sd}, \Theta_{sd}, \Gamma_{sd} \in \mathbb{R}^{n \times n}$, and $\psi_{sd} \in \mathbb{R}^n$.

By applying finite differences and being mindful of boundary conditions, matrix equations can be written for the solid and solution potentials.

$$\begin{bmatrix} \Lambda_{1,n} \phi_{1,n} & 0 \\ 0 & \Lambda_{1,p} \phi_{1,p} \end{bmatrix} - \begin{bmatrix} J_{1,n} + J_{sd} \\ J_{1,p} + B_{1,p} i_{app} \end{bmatrix} = 0 \quad (21)$$

$$\begin{bmatrix} \Lambda_{2,nm} & \Lambda_{2,ns} & \Lambda_{2,np} \\ \Lambda_{2,sn} & \Lambda_{2,ss} & \Lambda_{2,sp} \\ \Lambda_{2,pn} & \Lambda_{2,ps} & \Lambda_{2,pp} \end{bmatrix} \begin{bmatrix} \phi_{2,n} \\ \phi_{2,s} \\ \phi_{2,p} \end{bmatrix} + \begin{bmatrix} F_{c_2,n} \\ F_{c_2,s} \\ F_{c_2,p} \end{bmatrix} + \begin{bmatrix} J_{1,n} + J_{sd} \\ 0 \\ J_{1,p} \end{bmatrix} = 0 \quad (22)$$

where $\Lambda_{1,n} \in \mathbb{R}^{n \times n}$, $\Lambda_{1,p} \in \mathbb{R}^{p \times p}$, $B_{1,p} \in \mathbb{R}^p$, $\Lambda_{2,ij} \in \mathbb{R}^{i \times j}; i, j = n, p, s$ (the number of discretized points along the separator), and $[F_{c_2,n} \ F_{c_2,s} \ F_{c_2,p}]^T \in \mathbb{R}^{n+s+p}$ represents $\nabla \cdot (\kappa_D \nabla \ln(c_2))$.

Equations (18)-(22) form a system of linear algebraic equations, and can be combined into a single matrix equality. The solution to this matrix equality of form of $Ax = B$, is given by $x = A^{-1}B$. Several numerical simulations (provided in a later section) indicate successful inversion of this matrix under various loading conditions. The numerical complexity of the equation, however, depends on the number of discretization points along the electrodes and the separator. If n, s , and p indicate the number of points along anode, separator and cathode, respectively, A becomes a square matrix of size $4n+3p+s$. Although finer discretizations improve the simulation accuracy, they may significantly increase simulation time.

The quasi-linearization of intercalation current has several advantages. Firstly it replaces solving a set of nonlinear constraints with a set of linear constraints. Practically this trades an optimization problem with that of a linearization followed by solving a sparse set of linear equations. Secondly by recognizing that diffusion occurs much slower than electrical dynamics one realizes they can be treated as constant rather than variable, decreasing the number of items that need to be solved for. This decouples the ODEs and constraints at each time step allowing them to be solved in an alternating manner. Quasi-linearization reduces the electrochemical model by greatly reducing the numerical difficulties involved in satisfying the constraint equations. Reduction of the large number of states will be addressed in the following section.

A FAMILY OF PADÉ APPROXIMATIONS OF SPHERICAL DIFFUSION

If one has M points along the width of the cell, then roughly 2/3 of these points are in the anode and cathode (the rest would be

in the separator). The spherical diffusion submodel exists in the electrochemical model at every point along the width of the anode and cathode. If one has N states in the spherical diffusion model then the total number of states involved with spherical diffusion is $(2/3)M*N$. Put another way, lowering the number of states in the spherical diffusion model by one lowers the number of states in the full model by $(2/3)M$. Thus the importance of reducing this submodel becomes clear.

The diffusion model is a standard radially symmetric spherical diffusion model governed by Fick's law. The input to this system is either intercalation current or intercalation current density, both of which impose Neumann (slope) boundary conditions. The paper handles both cases at once since they differ only by a multiple. The system's two outputs are volume averaged total concentration $\bar{c}$ and surface concentration c_s . The total concentration $\bar{c}$ is related to battery SOC, and the surface concentration c_s affects the electrochemical reaction driving intercalation in the battery. The diffusion model governing these concentrations is a linear one-dimensional partial differential equation with one input and two outputs. It is given by:

$$\frac{\partial c(t, R)}{\partial t} = D \left(\frac{\partial^2 c(t, r)}{\partial r^2} + \frac{2}{r} \frac{\partial c(t, r)}{\partial r} \right), \quad (23)$$

$$\frac{\partial c(t, R)}{\partial r} = -mu(t), \quad (24)$$

$$\begin{bmatrix} c_s \\ \bar{c} \end{bmatrix} = \begin{bmatrix} c(t, R) \\ \left(\frac{\int_0^R c(t, r) (4\pi r^2 dr)}{(4/3)\pi R^3} \right) \end{bmatrix} \quad (25)$$

where $c(t, r)$ is the concentration at time t and radial distance r (where $r = 0$ is the center and $r = R$ is the surface), D is the diffusion coefficient, u is either an intercalation current leaving the sphere or an intercalation current density leaving the sphere, m is a factor relating this to the boundary slope, c_s is the concentration at the surface of the sphere and $\bar{c}$ is the total concentration per volume and R is the radius of the sphere.

One can compute transfer functions between the input and outputs by treating r as a parameter.

$$\mathcal{L} \left[\frac{\partial c(t, R)}{\partial t} \right] = \mathcal{L} \left[D \left(\frac{\partial^2 c(t, r)}{\partial r^2} + \frac{2}{r} \frac{\partial c(t, r)}{\partial r} \right) \right] \quad (26)$$

$$sC(s, r) = D \left(\frac{\partial^2 C(s, r)}{\partial r^2} + \frac{2}{r} \frac{\partial C(s, r)}{\partial r} \right) \quad (27)$$

Now consider the above one-dimensional boundary value problem with s as a parameter. This has the solution:

$$C(s, r) = \frac{e^{\sqrt{\frac{s}{D}}(R-r)} \left(e^{2\sqrt{\frac{s}{D}}r} - 1 \right) mR^2}{\left(1 + R\sqrt{\frac{s}{D}} + e^{2R\sqrt{\frac{s}{D}}} \left(R\sqrt{\frac{s}{D}} - 1 \right) \right) r} U(s) \quad (28)$$

Equation (28) provides a transfer function from the input to an arbitrary radial distance in the sphere. Jacobsen and West have a study on impedance involving spherical diffusion that makes nice use of transfer functions [17]. Applying the (linear) evaluation operator $r = R$ yields the transfer function for c_s :

$$\frac{C_s(s)}{U(s)} = - \frac{\left(e^{2R\sqrt{\frac{s}{D}}} - 1 \right) mR}{1 + R\sqrt{\frac{s}{D}} + e^{2R\sqrt{\frac{s}{D}}} \left(R\sqrt{\frac{s}{D}} - 1 \right)} \quad (29)$$

This is an infinite-dimensional transfer function with a countably infinite number of poles.

To evaluate the volume-averaged concentration, $\bar{c}$, one can use the following spherical integral:

$$\frac{\bar{C}(s)}{U(s)} = \frac{\int_0^R C(s, r) (4\pi r^2 dr)}{(4/3)\pi R^3} = - \frac{3Dm}{Rs} \quad (30)$$

This spherical integral has canceled an infinite number of poles leaving just a pure integrator. One need only integrate the input to compute $\bar{c}$ exactly. This makes physical sense if one considers a control volume around the sphere: one would only need to track what was going into and out of it to compute the total amount in the sphere. Since the averaged concentration transfer function is already quite simple in form we only need to make a Padé approximation for the surface concentration. This approximation is centered about zero due to the interest in lower-frequency behavior.

The Padé approximation is a rational polynomial approximation to a function centered about a point. One can choose the desired order of the numerator P and denominator Q . The Padé approximation matches the value of the approximated function and all of its derivatives up to $(P+Q)$ terms at the centered point. One can compute the Padé approximation by directly enforcing these constraints. Padé approximations are commonly used for time delay transfer functions, which are also infinite-dimensional. Padé approximations are a good choice for

approximating spherical diffusion because they can be centered about low frequency, handle infinite dimensionality without spatial discretization, are analytic, and can easily trade off speed versus accuracy by changing their order.

One slight modification to the standard Padé approximation is needed. Since the expansion is about zero and the function is undefined at zero (due to the removable pole at zero) first one multiplies by s and then computes the Padé approximation. After this we divide by s . This works because the pole at $s = 0$ is not an essential singularity. The order of the numerator is chosen one less than the order of the denominator, ensuring that the approximate transfer function is strictly proper.

The 2nd order Padé approximation for c_s will now be derived. The form of the Padé approximation is given in Eq. (31).

$$P(s) = \frac{a_0 + a_1 s}{s(1 + b_2 s)} \quad (31)$$

First we multiple Eq. (29) by s , and then we evaluate this quantity at $s = 0$. Then the same is done for the first and second derivative. This same procedure is performed on the Padé approximation and the derivatives of equal order are equated.

$$s \frac{C_s(s)}{U(s)} \Big|_{s=0} = sP(s) \Big|_{s=0} \Rightarrow -\frac{3dm}{R} = a_0 \quad (32)$$

$$\left(\frac{d}{ds} \left(s \frac{C_s(s)}{U(s)} \right) \right) \Big|_{s=0} = \left(\frac{d}{ds} (sP(s)) \right) \Big|_{s=0} \Rightarrow -\frac{mR}{5} = a_1 - a_0 b_2 \quad (33)$$

$$\left(\frac{d^2}{ds^2} \left(s \frac{C_s(s)}{U(s)} \right) \right) \Big|_{s=0} = \left(\frac{d^2}{ds^2} (sP(s)) \right) \Big|_{s=0} \Rightarrow \frac{2mR^3}{175} = 2b_2(-a_1 + a_0 b_2) \quad (34)$$

Solving these equations for a_0 , a_1 and b_2 one finds the approximation given in Eq. (31).

$$P(s) = -\frac{\frac{3dm}{R} + \frac{2mR}{7}s}{s + \frac{R^2}{35d}s^2} \quad (35)$$

The pole at origin in all of the Padé approximations can be used to compute $\bar{t}$ without increasing the number of states. One follows an identical method to derive higher order approximations.

The developed family of Padé approximations has several important characteristics. The family ranges from a one-state approximation to a countably infinite number of states, increasing in accuracy with the number of states. The approximations are generated through a systematic process with identical assumptions and all are analytic. The first few have been collected in Tab 1.

The 2nd to 20th order Padé approximations compare well in the frequency domain to the true model (Fig. 1). A comparison with finite differencing with 5 to 50 states in increments of 5 is shown Fig. 2, showing clearly that the Padé approximation provides much better accuracy per state.

TABLE 1. LOW ORDER PADÉ APPROXIMATIONS

Order	Padé Approximation
2 nd	$\frac{-\frac{3dm}{R} - \frac{2mR}{7}s}{s + \frac{R^2}{35d}s^2}$
3 rd	$\frac{-\frac{3Dm}{R} - \frac{4mRs}{11} - \frac{mR^3s^2}{165D}}{s + \frac{3R^2s^2}{55D} + \frac{R^4s^3}{3465D^2}}$
4 th	$\frac{-\frac{3Dm}{R} - \frac{2mRs}{5} - \frac{2mR^3s^2}{195D} - \frac{4mR^5s^3}{75075D^2}}{s + \frac{R^2s^2}{15D} + \frac{2R^4s^3}{2275D^2} + \frac{R^6s^4}{675675D^3}}$

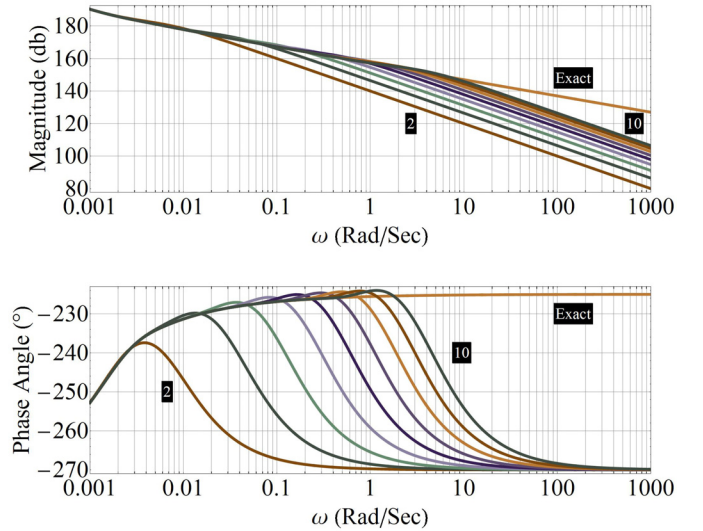


FIGURE 1. BODE PLOT OF PADÉ APPROXIMATION VERSUS TRUE MODEL

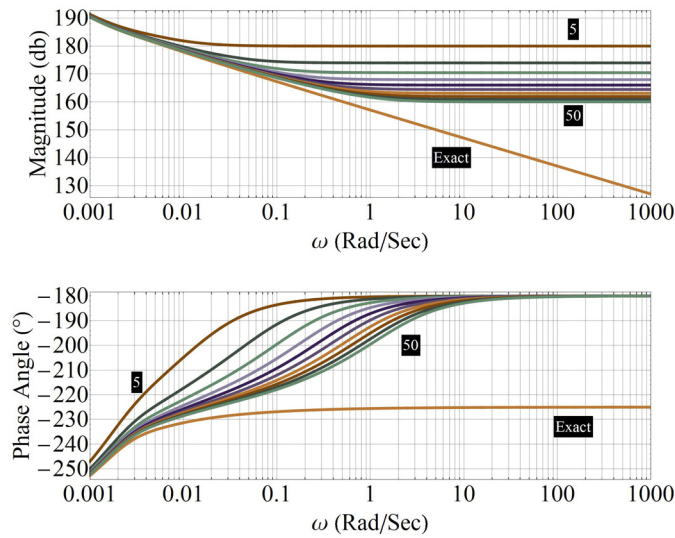


FIGURE 2. BODE PLOT OF FINITE DIFFERENCE METHOD VERSUS TRUE MODEL

COMPARISONS OF VARIOUS SPHERICAL DIFFUSION APPROXIMATIONS

The Padé approximation is not the first attempt at reducing the spherical diffusion model. When constrained to using two or fewer states the parabolic and quartic profile models are easy to implement and accurate at lower rates. When one has an idea of what signals will be fed into the model POD is a good choice. For a specific frequency range window Residue Grouping (RG) is very effective. Both POD and RG are numerical in nature. Padé approximation allows one to have an analytical method with an arbitrary number of states. This section presents comparisons in the frequency domain of the Padé approximation and various other approximations.

This paper does not nondimensionalize the spherical diffusion models because the main interest is how well they work inside the full battery model. A representative set of parameters is used for comparing the approximations. The parameters are given in Tab. 2 and are all identical to those in both [10-11]. This makes the values in the Bode plots more physically meaningful.

TABLE 2. SPHEREICAL DIFFUSION PARAMETERS

Parameter	Value
D (m ² /sec)	$2 \cdot 10^{-16}$
R (m)	$1 \cdot 10^{-6}$
m (sec/m ²)	$1/D$

While the single-input two-output spherical diffusion system has two transfer functions all the Bode plots are of c_s . Bode plots of $\bar{c}$ would not be enlightening since $\bar{c}$ is a pure integrator and

every approximation with a zero eigenvalue will match it perfectly.

Parabolic and Quartic Profile Approximations

In paper [9], two ways of approximating spherical diffusion are considered. Both are based on approximating the spherical profile. One assumes a parabolic profile, and the other assumes a quartic profile. The surface concentration transfer functions for each are given in Eqs. (36) and (37) respectively.

$$-\frac{15mD - R^2ms}{5Rs} \quad (36)$$

$$-\frac{3150D^2m + 315DmR^2 + mR^4s^2}{35Rs(30D + R^2s)} \quad (37)$$

Bode plots of these along with the exact transfer function are presented in Fig. 3.

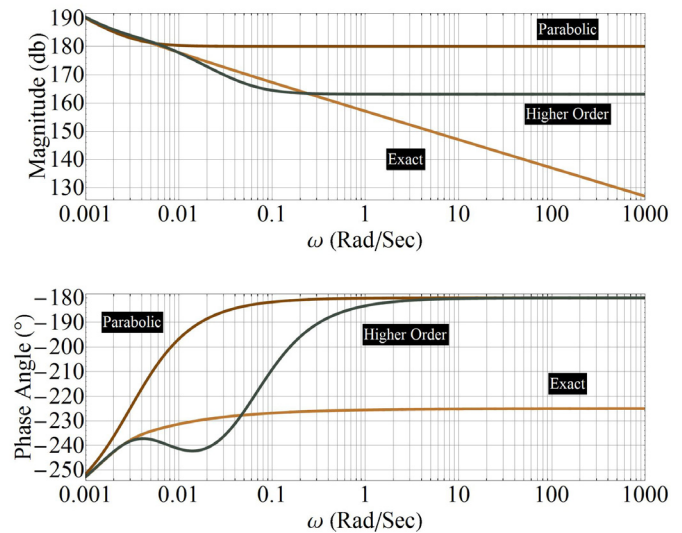


FIGURE 3. BODE PLOT OF THE PARABOLIC PROFILE AND QUARTIC PROFILE MODELS VERSUS THE TRUE MODEL

The parabolic profile has been shown to be effective in low rate battery simulation [12]. However they lack a scheme to generalize to higher orders. The higher-order polynomial which has two states slightly outperforms the second order Padé approximation. This is due to the higher model having a direct input, whereas the Padé approximations are all strictly proper. Interestingly enough a non-strictly proper 1st order Padé approximation is identical to the parabolic approximation.

POD Approximation

Paper [10] presents a POD based model reduction of spherical diffusion. The Bode plot of this and Padé approximation 6, 7 and 12 are given in Fig. 4.

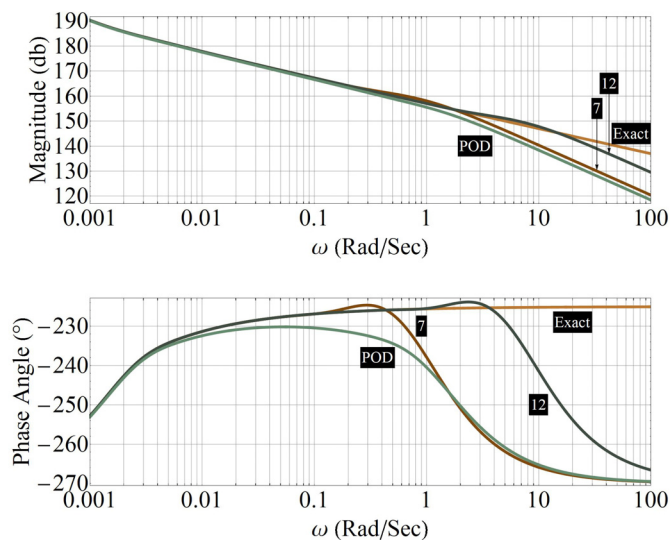


FIGURE 4. BODE PLOT OF POD, VARIOUS PADÉ APPROXIMATION AND THE TRUE MODEL

The approximation given in the paper is 12th order, a 7th order Padé approximation exceeds its accuracy. The 12th order is given to compare the responses with an equal number of states. An advantage of the POD method is that it allows the reconstruction of the entire state of the system, whereas the Padé approximation does not.

Residue Grouping

Residue Grouping can also be used to reduce the diffusion models [11]. Instead of approximating the surface concentration, this paper approximates the difference of the surface concentration to the average concentration. Thus the correct steady state behavior is zero, so there is no need to have a zero eigenvalue. This method focuses on matching the model in a specific frequency range. This is different from the Padé approximation as it matches steady state exactly and then extends towards faster frequencies. In moving from a Padé approximation to a residue grouping approximation one gives up some accuracy at lower frequencies, but in return one can get a good approximation for a wider frequency range. Fig. 5 is of Padé approximations to the surface concentration minus the average concentration.

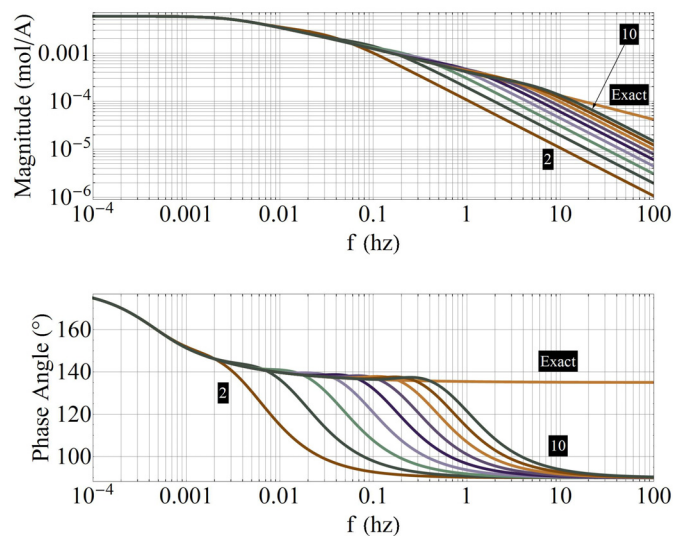


FIGURE 5. BODE PLOT PADÉ APPROXIMATION FOR COMPARISON WITH FIGURE 3 PAPER [11]

The Bode plot is in hertz, and the magnitude values on the magnitude plot are absolute, this makes comparison easier to Figure 3 in [11].

Summary: Advantages of Padé Approximation

The family of Padé approximations is analytic, generalizable to any order, does not involve discretization and preserve steady state behavior. They compare favorably with finite differencing, parabolic and higher order polynomial approximations and POD. While Padé approximations are better than Residue Grouping at very low frequencies, Residue Grouping is an excellent method if one is concerned with a frequency window.

NUMERICAL SIMULATIONS

Numerical simulation of the model with constraint quasi-linearization and Padé approximation has been carried out to evaluate how the methods affect model accuracy and computational speed. Model parameters are identical to those in [7]. 20 hours worth of constant current constant voltage (CCCV) cycles were simulated using 1A (16.54 A/m² density) of current and voltage limits of 3.1V and 4.2V.

While Padé approximation and finite difference both converge to accurate solutions - the Padé approximation converges at a much quicker rate with model order, as seen in Fig. 6.

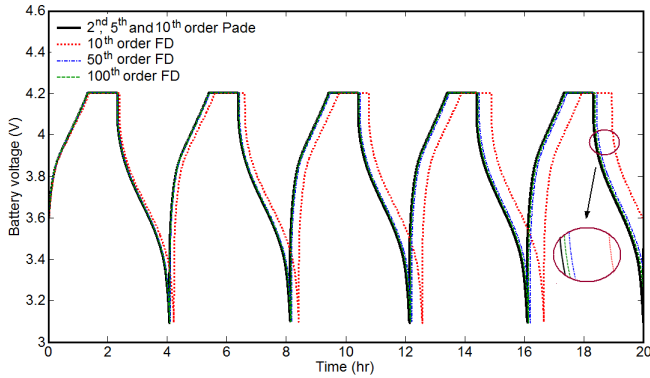


FIGURE 6. SIMULATION OF FULL MODEL WITH VARIOUS PADÉ APPROXIMATIONS AND FINITE DIFFERENCES

One achieves very high accuracy with a 10th order Padé approximation whereas one needs a 100th order finite difference model, for the CCCV cycling. Here one can reduce the number of state variables in each submodel by 90 and thus the full model by approximately $90 \cdot (2/3) \cdot M$ state variables. For $M = 100$ this results in a model with 6700 state variables decreasing to one with 670. This can lead to large increases in numerical performance. For example, on an implementation in MATLAB on a computer with a 2.0 GHz processor using 10 second time steps it takes 435 seconds to run with a 100 order finite difference model. However with the 10th order Padé model this same simulation takes 75 seconds. This comparative time benefit is solely due to the Padé approximation as in both cases quasi-linearization was used. Being able to compute 20 hours of simulation in 75 seconds is 3 orders of magnitude faster than real time and makes doing degradation studies more tractable. The advantage is especially important for optimization where one may be simulating the battery for several hours for each function call the optimizer uses.

Clearly different implementations will have varying times. This makes it important to understand the improvement in theoretical performance. Since both systems are constant and linear one can use the state transition matrix to compute exact solutions of each. In simulation this would be done at every time step and would be dominated by the matrix multiplication of the state transition matrix. This matrix vector multiplication takes time $O(N^2)$, thus reducing the number of state variables has a quadratic affect on the Big O notation time[19]. If one is only concerned with the spherical diffusion submodel and the state variables are reduced by a factor of 10 one would expect a 100 fold improvement in the submodel runtime. Padé approximation is a powerful tool for reducing both the number of state variables and computational time of the full model.

Quasi-linearization is a highly accurate way to solve the nonlinear constraint equations. The natural way to check this is to compute how accurately the quasi-linear solution satisfies the nonlinear constraints. Since there are a large number of constraints at each time step it is best to use two relative error norms to make sense of the data. The maximum relative error is

given by an L^∞ norm in space and the average relative error is given by an L^1 norm in space. Fig. 7 shows both of these error norms during simulation and neither is ever greater than 0.01%. In fact, for the vast majority of the time, they are much lower.

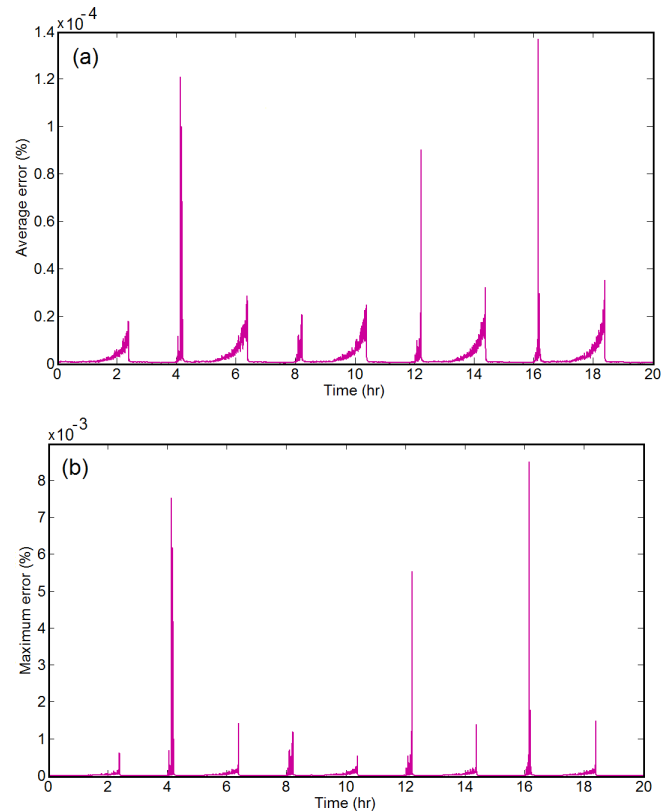


FIGURE 7. RELATIVE ERRORS DURING SIMULATION

It is difficult to determine how much faster quasi-linearization is over other methods because there are a vast number of methods for solving nonlinear equations. Quasi-linearization does have the computational advantage that it does not have to iterate repeatedly at each time step unlike many equation solving methods. It is the author's experience that quasi-linearization is substantially faster than using optimization to solve the nonlinear constraint equations. Applying both the Padé approximation and constraint quasi-linearization greatly reduce the computational complexity of the model while maintaining very high accuracy.

Conclusion

Constraint quasi-linearization and Padé approximation of diffusion substantially decrease the complexity and numerical cost of this electrochemical model. This decrease allows the model to be used for studying long term battery health degradation. Preliminary studies using this model indicate that battery SOC has an essential role in battery health degradation.

ACKNOWLEDGMENTS

The financial support of the NSF EFRI-RESIN program and the Michigan Public Service Commission is gratefully acknowledged.

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